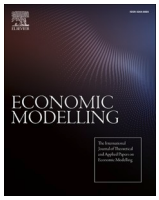
Economic Modelling 123 (2023) 106287



Contents lists available at ScienceDirect

Economic Modelling

journal homepage: www.journals.elsevier.com/economic-modelling





Sequential Bayesian analysis for semiparametric stochastic volatility model with applications☆

Nianling Wang [a,*], Zhusheng Lou [b,**]

[a] School of Finance, Capital University of Economics and Business, China
[b] China Construction Bank Corporation, China

ARTICLE INFO

Handling Editor: Sushanta Mallick

Keywords:
Stochastic volatility model
Spline
Sequential Monte Carlo
Particle Markov chain Monte Carlo

ABSTRACT

The stochastic volatility (SV) model is widely used to study time-varying volatility. However, the linearity assumption for transition equation in basic SV model is restrictive. To allow for nonlinearity, we proposed a semiparametric SV model that specifies a nonparametric transition equation for log-volatility using natural cubic splines. To estimate the semiparametric SV model, we used the sequential Monte Carlo algorithm and particle Markov chain Monte Carlo methods, which are shown to be able to provide effective estimates of the model in simulation studies. The empirical applications to Bitcoin and convertible bond return data indicate that the transition equations of their log-volatility are highly nonlinear. Taking nonlinearity into account, the semiparametric SV model can improve the likelihood of the basic SV model both in-sample and out-of-sample. Furthermore, the semiparametric SV model produces more flexible estimated volatility and can better filter out volatility clustering characteristics in original return data.

1. Introduction

Volatility is a major concern in finance because it affects many financial activities, such as asset pricing, portfolio construction, risk management, and hedging. The volatility of asset returns is time-varying; hence, various models are proposed for volatility modeling and prediction, among which stochastic volatility (SV) models are powerful and flexible (Kim et al., 1998; Yu, 2002). The SV models first appeared in studies on option pricing as a modification to the Black–Scholes model (Hull and White, 1987), and they have since been empirically formulated in discrete-time version to model dynamic volatility and are widely used in finance (Yu et al., 2006; Xu et al., 2011; Zeng and Taksar, 2013; Huang, 2015; Chen et al., 2017; Tiwari et al., 2019; Hafner, 2020; Cross et al., 2021; Ulm and Hambuckers, 2022).

In the transition equation, the basic SV model assumes Gaussian innovation and linearity. This facile specification is used for ease of estimation; however, it limits the flexibility and generality of the model. Further, different variants of the basic SV model were proposed, such as SV models with a leverage effect (Yu, 2005; Li et al., 2012), SV models with non-Gaussian innovations (Jensen and Maheu, 2010; Delatola and Griffin, 2011), SV models with Markov switching (Pan and Li, 2013; He and Lin, 2023), and semiparametric SV models (Yu, 2012; Langrock et al., 2015; Hu et al., 2021).

The current literature on semiparametric SV models is primarily concerned with the distribution of innovation. Jensen and Maheu (2010) introduced a nonparametric Dirichlet process mixture of innovations to propose a flexible semiparametric SV model. Delatola and Griffin (2011) established that the error term distribution follows an infinite mixture of normal via the Dirichlet process mixture models. Yu (2012) developed a semiparametric SV model whose correlation between return and volatility innovation is time-varying. Langrock et al. (2015) found that the error term distribution is nonparametric and estimated the conditional distribution via the B-spline method.

Unlike previous studies, we are particularly interested in the functional form of the transition equation in the SV model, which is usually restricted to being a linear $AR(1)$ process for simplicity. Several studies have noted this unreasonable assumption and proposed nonlinear SV models in various ways. Yu et al. (2006) proposed a nonlinear SV model

☆ The authors would like to thank Sushanta Mallick (the editor), Yong Li (the guest editor) and two referees for their valuable comments, which helped a lot to improve the quality of the paper. Wang gratefully acknowledges the financial support of the Chinese Natural Science Fund (No.72203151) and the support by Capital University of Economics and Business (No. XRZ2022037).

* Corresponding author.
** Corresponding author
E-mail addresses: nianling.wang@cueb.edu.cn (N. Wang), louzhusheng@163.com (Z. Lou).

https://doi.org/10.1016/j.econmod.2023.106287
Received 29 December 2022; Received in revised form 17 March 2023; Accepted 18 March 2023
Available online 22 March 2023
0264-9993/© 2023 Elsevier B.V. All rights reserved.

using the Box–Cox transformation to the volatility transition equation. Hu et al. (2021) proposed a nonlinear SV model with a semiparametric conditional variance equation, estimated using kernel-form densities. He and Lin (2023) introduced regime switching to the long-run mean of volatility to account for the economic cycle condition in this nonlinear SV model.

In this study, we modeled the transition equation in nonparametric form using the natural cubic spline to relax the linearity assumption in a simple but efficient manner. Because the variables in the transition equation are unknown, determining the domain of the spline is difficult. Based on the property of the natural cubic spline at two domain endpoints, we extended the original domain to a real number set and introduced linear relationship outside its original domain. The nonparametric specification allows for more flexibility in the log-volatility transition equation. The proposed model is shown to provide a better fit of data in both our simulation study and empirical applications.

For SV models, the main difficulty is estimation, particularly with non-Gaussian and nonlinear specifications. Because the likelihood is difficult to evaluate, applying the traditional estimation procedure directly is difficult. For parameter estimation in SV model studies, the quasi-maximum likelihood and Markov chain Monte Carlo (MCMC) methods are proposed (Harvey et al., 1994; Jacquier et al., 1994; Kim et al., 1998; Sandmann and Koopman, 1998; Yu et al., 2006; Hu et al., 2021).

With the development of sequential Monte Carlo (SMC), a particle filter serves as a flexible tool to estimate the SV model. Particle filter was first proposed to estimate the likelihood in SV models by Kim et al. (1998). Based on the particle algorithm and asymptotic convergence of the MCMC algorithm, Andrieu et al. (2010) proposed the particle MCMC (PMCMC) method that combines and takes advantage of both the SMC and MCMC methods. PMCMC extends the traditional MCMC algorithm to more general settings, such as non-Gaussian and nonlinear SV model specifications, and it can give the estimate of Bayes factor for model comparison as a by-product. Li et al. (2022) employed PMCMC method for bandwidth selection in kernel regression. In this study, we used the SMC algorithm and PMCMC method to estimate the proposed semiparametric SV model. The simulation results demonstrate that the algorithm can provide a robust and effective estimation for our model.

For empirical applications, we applied our model to two financial products, Bitcoin and a convertible bond for Shanghai Modern Pharmaceutical Company Limited. Because Bitcoin is often used for speculation (Baek and Elbeck, 2015; Dyhrberg, 2016; Hafner, 2020), it is highly volatile with a complicated volatility process. Because the convertible bond contains the option to convert to stock at a convertible price, the pricing principal varies based on the relative movement of the stock and convertible prices. When stock price is lower, close to, or higher than the convertible price, the convertible bond behaves similarly to an ordinary bond, an option, or a stock, respectively. Therefore, we expect some nonlinearity in the log-volatility transition equations of these two financial products. We investigated this nonlinearity using our proposed semiparametric SV model and compared the results to those obtained using the basic SV model. The empirical results show that there is actually high nonlinearity in the log-volatility transition equation. The slope of the transition function is not a constant as assumed by the basic SV model but varies greatly, ranging from −0.25 (−0.75) to 1.75 (2.75) for Bitcoin (convertible bond) return data. Taking nonlinearity into account, our semiparametric SV model can improve the likelihood of the basic SV model both in-sample and out-of-sample. Moreover, compared to the basic SV model, the semiparametric SV model obtains a more flexible estimate of volatility following volatility shocks and can better filter out the volatility clustering characteristics in the original return data. These two empirical examples highlight the importance of using this flexible semiparametric SV model to characterize the nonlinearity of log-volatility transition in real-world data.

This study makes three major contributions. First, for the volatility modeling literature, we focused on the log-volatility transition process and relaxed the linearity assumption in a nonparametric way. We proposed a semiparametric SV model in which a natural cubic spline function is used to approximate a flexible nonparametric transition process, with the transition equation incorporating any functional relationship. Second, for the estimation method, we used the SMC algorithm and PMCMC method to estimate the proposed semiparametric SV model. The simulation results show that the algorithm provides a robust and effective estimation of the model. Third, we applied our model and method to Bitcoin and convertible bond return data, and empirically found that there is actually high nonlinearity in the transition equations of log-volatility. Taking the nonlinearity into account can improve the likelihood of basic SV model both in-sample and out-of-sample.

The paper is organized as follows: Section 2 introduces the natural cubic spline and presents a semiparametric SV model. Section 3 introduces the SMC algorithm and the PMCMC method for estimating the proposed model. Sections 4 and 5 present one simulation study and two empirical applications, respectively. Section 6 concludes the paper.

2. Semiparametric SV model

2.1. Natural cubic spline

First, we briefly introduce the natural cubic spline used for nonparametric modeling of the transition equation in the semiparametric SV models proposed later. The cubic spline function is a piecewise polynomial with the order cubic predetermined. A typical cubic spline, denoted by g, takes values from the interval $[a, b]$ and maps them to the set of real numbers.

$$g : [a, b] \rightarrow \mathbb{R} \tag{1}$$

The flexibility of the cubic spline function is determined by the number of knots in its domain. For a typical cubic spline g, it usually has $K + 1$ knots on the domain satisfying,

$$\begin{aligned}[a, b] = [t_0, t_1] \cup [t_1, t_2] \cup \ldots \cup [t_{K-1}, t_K],\\ a = t_0 < t_2 < \ldots < t_K = b.\end{aligned} \tag{2}$$

Usually, the knots are spaced evenly. Moreover, to impose smoothness, it is common to assume that the cubic spline function is continuous, first-order, and second-order continuously differentiable at each knot. Let $g_i(.)$ represent the cubic function in the interval $[t_{i-1}, t_i]$, and it satisfies,

$$g_i(t_i) = g_{i+1}(t_i), \quad g_i'(t_i) = g_{i+1}'(t_i), \quad g_i''(t_i) = g_{i+1}''(t_i). \tag{3}$$

Particularly, for the spline-based semiparametric SV model proposed in the following subsection, it is difficult to determine the interval $[a, b]$ that can cover all possible values of independent variables because they are unobservable and sensitive to model parameters via the particle filter.

To address this issue, we used the natural cubic spline function instead of the traditional cubic spline function. For the natural cubic spline, the second derivatives of the spline cubic polynomials were assumed to be zero at the endpoints of the interval $[a, b]$.

$$g''(a) = g''(b) = 0 \tag{4}$$

Thus, at the endpoints of the natural cubic spline, we performed a linear approximation through its first derivatives. The function outside the interval $[a, b]$ was a linear function determined using the value at the endpoint of the interval and its first derivatives. In practice, we can approximate the interval outside $[a, b]$ and avoid the issue that the domain of the natural cubic spline cannot contain all possible values of independent variables. Based on the natural cubic spline, we extended the interval $[a, b]$ to the set of real numbers.

Let $\boldsymbol{g} = (g_0, g_1, \ldots, g_K)^T$ be a column vector of values $g_i = g(t_i)$, $i =$

$0, 1, \ldots, K$. Green and Silverman (1993) established that, given any values of $g_0, g_1, \ldots, g_K$, there is a unique natural cubic spline. Therefore, during the estimation, given the interval $[a, b]$ and knots, we can equivalently transfer the natural cubic spline to a parameter vector $\boldsymbol{g}$.

2.2. SV model with a nonparametric transition equation

The SV model is a typical state space model comprising a measurement equation and a transition equation. Let y_t be a financial time series with a time-varying log-volatility h_t. The measurement equation connects the observed data y_t and unobserved state variable h_t. The transition equation describes the process of the unobserved state variable h_t. For the basic SV model, the transition equation is often assumed to be an $AR(1)$ process and expressed as follows:

$$\begin{aligned} y_t &= \alpha + \exp\left(\frac{h_t}{2}\right) v_t \quad , \\ h_{t+1} &= \mu(1-\varphi) + \varphi h_t + \sigma w_t \quad , \end{aligned} \tag{5}$$

where v_t and w_t are the independent and identically distributed random variables following a standard normal distribution.

However, the real economic relationship in the transition equation is unlikely to be a simple linear $AR(1)$ process. The linearity assumption in the basic SV model is strong and may be detrimental to the empirical performance of the model. Intuitively, a nonlinear transition function would work better with real data and satisfy the realistic requirements of SV modeling.

Herein, we present a flexible spline-based nonparametric transition process in the SV model. The semiparametric SV model is expressed as follows:

$$\begin{aligned} y_t &= \alpha + \exp\left(\frac{h_t}{2}\right) v_t \quad , \\ h_{t+1} &= m_h(h_t) + \sigma w_t \quad , \end{aligned} \tag{6}$$

where $m_h(\bullet)$ is a smooth but unknown function. During the estimation, a natural cubic spline is applied to estimate $m_h(\bullet)$.

In the semiparametric SV model, the conditional probability distributions for observations y_t and state variable h_t are expressed as follows:

$$\begin{aligned} p(y_t|h_t;\theta) &= (2\pi e^{h_t})^{-\frac{1}{2}} \exp\left(-\frac{(y_t-\alpha)^2}{2e^{h_t}}\right), \\ p(h_{t+1}|h_t;\theta) &= (2\pi\sigma^2)^{-\frac{1}{2}} \exp\left(-\frac{(h_{t+1}-m_h(h_t))^2}{2\sigma^2}\right). \end{aligned} \tag{7}$$

3. Sequential Monte Carlo algorithm

3.1. Particle filter

Let $y_{1:t} = \{y_1, \ldots, y_t\}$ be the observed data and $h_{0:t} = \{h_0, \ldots, h_t\}$ be the unobserved log-volatility. For a particle filter, the joint smoothing distribution $p(h_{0:t}|y_{1:t};\theta)$ is the key to applying the sampling method. Calculating the joint smoothing distribution and directly sampling from it are difficult. However, it can be expressed as follows:

$$\begin{aligned} p(h_{0:t}|y_{1:t};\theta) &= \frac{p(y_t|h_{0:t}, y_{1:t-1};\theta)p(h_{0:t}|y_{1:t-1};\theta)}{p(y_t|y_{1:t-1};\theta)} \\ &= \frac{p(y_t|h_{0:t}, y_{1:t-1};\theta)p(h_t|h_{0:t-1}, y_{1:t-1};\theta)p(h_{0:t-1}|y_{1:t-1};\theta)}{p(y_t|y_{1:t-1};\theta)} \\ &= \frac{p(y_t|h_t;\theta)p(h_t|h_{t-1};\theta)}{p(y_t|y_{1:t-1};\theta)} p(h_{0:t-1}|y_{1:t-1};\theta). \end{aligned} \tag{8}$$

The sampling method can be implemented using the recursive form. The particle filter samples log-volatility by using one-step-ahead prediction and filtering recursively. Based on the data sampled at time $t-1$, the one-step prediction is formed as follows:

$$p(h_t|y_{1:t-1};\theta) = \int p(h_t|h_{t-1};\theta)p(h_{t-1}|y_{1:t-1};\theta)dh_{t-1} \ . \tag{9}$$

When there is a new observation y_t, the updating step is applied to obtain the filtering distribution by using Bayes' rule,

$$\begin{aligned} p(h_t|y_{1:t};\theta) &= \frac{p(y_t, h_t|y_{1:t-1};\theta)}{p(y_t|y_{1:t-1};\theta)} \\ &= \frac{p(y_t|h_t;\theta)p(h_t|h_{t-1};\theta)p(h_{t-1}|y_{1:t-1};\theta)}{\int p(y_t|h_t;\theta)p(h_t|h_{t-1};\theta)p(h_{t-1}|y_{1:t-1};\theta)dh_t}. \end{aligned} \tag{10}$$

Because the target joint smoothing distribution is often not a standard distribution from which direct draws are difficult, a proposal distribution $g_{0:t}(h_{0:t}|y_{1:t};\theta)$ is employed to perform importance sampling. If the proposal distribution can be factored into two parts as shown below:

$$g_{0:t}(h_{0:t}|y_{1:t};\theta) = g_t(h_t|h_{0:t-1}, y_{1:t};\theta)g_{0:t-1}(h_{0:t-1}|y_{1:t};\theta). \tag{11}$$

Then the importance weights of sample $\{h_{0:t}^{(i)}\}_{i=1:M}$ satisfy

$$\begin{aligned} \tilde{w}_t^{(i)} &\propto \frac{p(h_{0:t}|y_{1:t};\theta)}{g_{0:t}(h_{0:t}|y_{1:t};\theta)} \\ &= \frac{p(y_t|h_t;\theta)p(h_t|h_{t-1};\theta)}{p(y_t|y_{1:t-1};\theta)g_t(h_t|h_{0:t-1}, y_{1:t};\theta)} \frac{p(h_{0:t-1}|y_{1:t-1};\theta)}{g_{0:t-1}(h_{0:t-1}|y_{1:t};\theta)} \\ &= \frac{p(y_t|h_t;\theta)p(h_t|h_{t-1};\theta)}{p(y_t|y_{1:t-1};\theta)g_t(h_t|h_{0:t-1}, y_{1:t};\theta)} \tilde{w}_{t-1}^{(i)} \\ &\propto \frac{p(y_t|h_t;\theta)p(h_t|h_{t-1};\theta)}{g_t(h_t|h_{0:t-1}, y_{1:t};\theta)} \tilde{w}_{t-1}^{(i)} \ . \end{aligned} \tag{12}$$

Particularly, the proposal distribution $g_t(h_t|h_{0:t-1}, y_{1:t};\theta)$ can be set as $p(h_t|h_{t-1};\theta)$ directly, and it is easy to combine the importance sampling with the prediction and filtering steps. The weights in the importance sampling can be updated in the recursive sampling.

In addition to the posterior sampling of log-volatility, the marginal density is also an important output of the particle filter. Under the recursive importance sampling method, the marginal density can be written as follows:

$$\begin{aligned} p(y_t|y_{1:t-1};\theta) &= \int p(y_t|h_t;\theta)p(h_{0:t}|y_{1:t-1};\theta)dh_{0:t} \\ &= \int \frac{p(y_t|h_t;\theta)p(h_t|h_{t-1};\theta)}{g_t(h_t|h_{0:t-1}, y_{1:t};\theta)} \frac{p(h_{0:t-1}|y_{1:t-1};\theta)}{g_{0:t-1}(h_{0:t-1}|y_{1:t};\theta)} g_{0:t}(h_{0:t}|y_{1:t};\theta)dh_{0:t} \ . \end{aligned} \tag{13}$$

Let $\{h_{0:t-1}^{(i)}, \widetilde{w}_{t-1}^{(i)}\}_{i=1:M}$ be the posterior samples with importance weights at time $t-1$ and $\{h_t^{(i)}\}_{i=1:M}$ be the samples from the proposal distribution at time t. The marginal density can be calculated as follows:

$$p(y_t|y_{1:t-1};\theta) = \sum_{i=1}^{M} \frac{p\left(y_t|h_t^{(i)};\theta\right)p\left(h_t^{(i)}\middle|h_{t-1}^{(i)};\theta\right)}{g_t\left(h_t^{(i)}\middle|h_{0:t-1}^{(i)}, y_{1:t};\theta\right)} \tilde{w}_{t-1}^{(i)}. \tag{14}$$

Therefore, the particle filter can be implemented to calculate the likelihood. Particularly, for the state space model with a linear relationship and Gaussian distribution, the particle filter is the same as the Kalman filter. Hence, the particle filter is more general and can be used for state space models with nonlinear relationships and non-Gaussian distributions, such as the semiparametric SV model proposed herein. The particle filter used is summarized as follows.

Particle filter

Initialization

Given θ and particle number M, draw the distribution $\left\{h_0^{(i)}, w_0^{(i)} = \frac{1}{M}\right\}_{i=1:M}$.

For $t = 1, \ldots, T$, do

1. Given the distribution $\left\{h_{t-1}^{(i)}, \frac{1}{M}\right\}_{i=1:M}$, draw current particle $\widetilde{h}_t^{(i)}$ in the proposal $p\left(h_t\middle|h_{t-1}^{(i)};\theta\right)$ with weights $\widetilde{w}_t^{(i)} \propto \frac{1}{M} \times p\left(y_t\middle|\widetilde{h}_t^{(i)};\theta\right)$.
2. Calculate the marginal density $p(y_t|y_{1:t-1};\theta) \approx \sum_{i=1}^{M} \widetilde{w}_t^{(i)}$.

(*continued on next page*)

(*continued*)

Particle filter

3. Build $\left\{\widetilde{h}_t^{(i)}, \frac{\widetilde{w}_t^{(i)}}{\sum_{i=1}^{M}\widetilde{w}_t^{(i)}}\right\}_{i=1:M}$ and smooth resample M particles to obtain distribution $\{h_t^{(i)}, \frac{1}{M}\}_{i=1:M}$.
4. Calculate the estimate log-volatility $\widehat{h}_t = \frac{1}{M}\times\sum_{i=1}^{M}h_t^{(i)}$.

end for $\log(p(y_{1:T}|\theta)) = \sum_{t=1}^{T}\log(p(y_t|y_{1:t-1};\theta))$

3.2. Particle Markov chain Monte Carlo

The PMCMC algorithm includes the particle independent Metropolis–Hastings sampler (PIMH), the particle marginal Metropolis–Hastings sampler (PMMH), and the particle Gibbs sampler. In this study, we used the PMMH algorithm to estimate the semiparametric SV model, which updates the parameter and state variables jointly.

Assume that the sampling from conditional density $p(h_{1:T}|y_{1:T};\theta)$ is feasible and that the standard decomposition is

$$p(\theta, h_{1:T}|y_{1:T}) = p(\theta|y_{1:T})p(h_{1:T}|y_{1:T};\theta). \tag{15}$$

Therefore, the proposal distribution for MH updates can be

$$q\{(\theta^*, h_{1:T}^*)|(\theta, h_{1:T})\} = q(\theta^*|\theta)\, p(h_{1:T}^*|y_{1:T};\theta^*). \tag{16}$$

In the above proposal distribution, $p(h_{1:T}^*|y_{1:T};\theta^*)$ can be implemented through a particle filter directly; Therefore, the only freedom in this algorithm is the proposal distribution $q(\theta^*|\theta)$. The marginal density $p(\theta|y_{1:T})$ also satisfies

$$p(\theta|y_{1:T}) \propto p(\theta)p(y_{1:T}|\theta). \tag{17}$$

The MH acceptance ratio is expressed as follows:

$$\begin{aligned}\alpha((\theta, h_{1:T}), (\theta^*, h_{1:T}^*)) &= \frac{p(\theta^*, h_{1:T}^*|y_{1:T})q\{(\theta, h_{1:T})|(\theta^*, h_{1:T}^*)\}}{p(\theta, h_{1:T}|y_{1:T})q\{(\theta^*, h_{1:T}^*)|(\theta, h_{1:T})\}} \\ &= \frac{p(\theta^*|y_{1:T})p(h_{1:T}^*|y_{1:T};\theta^*)q(\theta|\theta^*)\, p(h_{1:T}|y_{1:T};\theta)}{p(\theta|y_{1:T})p(h_{1:T}|y_{1:T};\theta)q(\theta^*|\theta)\, p(h_{1:T}^*|y_{1:T};\theta^*)} \\ &= \frac{p(\theta^*)p(y_{1:T}|\theta^*)q(\theta|\theta^*)}{p(\theta)p(y_{1:T}|\theta)q(\theta^*|\theta)}.\end{aligned} \tag{18}$$

In this study, the PMMH algorithm implemented is summarized as follows:

Particle Marginal Metropolis–Hastings

Initialization

1. An initial parameter value $\theta^{(0)}$, a proposal distribution $q(\theta^*|\theta)$, the number of particles M, the number of burn-in iterations B, and the number of MCMC iterations N.
2. Conditioning on $\theta^{(0)}$, calculate the state variable $h_{1:T}^{(0)}$ and $p(y_{1:T}|\theta^{(0)})$ using the particle filter method

For $i = 1,\ldots,(B+N)$, do

1. Sample θ' from $q(\theta|\theta^{(i-1)})$.
2. Conditioning on θ', calculate the $h_{1:T}'$ and $p(y_{1:T}|\theta')$ using the particle filter method.
3. With probability $\min\left\{\frac{p(\theta^*)p(y_{1:T}|\theta^*)q(\theta|\theta^*)}{p(\theta)p(y_{1:T}|\theta)q(\theta^*|\theta)}, 1\right\}$.

set $\theta^{(i)} = \theta'$ and $h_{1:T}^{(i)} = h_{1:T}'$; otherwise, set $\theta^{(i)} = \theta^{(i-1)}$ and $h_{1:T}^{(i)} = h_{1:T}^{(i-1)}$

end

4. Simulation study

4.1. The prior and proposal distribution

For the parameter in the semiparametric SV model, we set the prior to be,

$$\alpha, \beta, \varphi, \log(\sigma^2) \sim N(0, 100), \tag{19}$$

and the proposal distribution to be,

$$\theta^*|\,\theta^{(i-1)} \sim N(\theta^{(i-1)}, \sigma_e^2), \tag{20}$$

where the constant σ_e^2 is specified by users in specific applications.

For the parameter vector $\boldsymbol{g}$ in natural cubic spline, Green and Silverman (1993) demonstrated that the roughness penalty satisfies,

$$\int g''(t)^2 dt = \boldsymbol{g}^T K_{ncs}\boldsymbol{g}, \tag{21}$$

where $K_{ncs} = QR^{-1}Q^T$. Define $h_i = t_{i+1} - t_i$, $i = 0, 1, \ldots, K-1$, matrices Q and R satisfy

$$Q = \begin{bmatrix} 1/h_0 & 0 & 0 & \cdots & 0 \\ -\frac{1}{h_0}-\frac{1}{h_1} & \frac{1}{h_1} & 0 & \cdots & 0 \\ \frac{1}{h_1} & -\frac{1}{h_1}-\frac{1}{h_2} & \frac{1}{h_1} & \cdots & 0 \\ 0 & \frac{1}{h_2} & -\frac{1}{h_2}-\frac{1}{h_3} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & -\frac{1}{h_{K-2}}-\frac{1}{h_{K-1}} \\ 0 & 0 & 0 & \cdots & 1/h_{K-1} \end{bmatrix}_{(K+1)\times(K-1)},$$

$$R = \begin{bmatrix} \frac{h_0+h_1}{3} & \frac{h_1}{6} & 0 & \cdots & 0 \\ \frac{h_1}{6} & \frac{h_1+h_2}{3} & \frac{h_2}{6} & \cdots & 0 \\ 0 & \frac{h_2}{6} & \frac{h_2+h_3}{3} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \frac{h_{K-2}+h_{K-1}}{3} \end{bmatrix}_{(K-1)\times(K-1)}. \tag{22}$$

Green and Silverman (1993) established that the prior log density for the natural cubic spline could satisfy,

$$l_{prior}(g) = -\frac{1}{2}\lambda\int g''(t)^2 = -\frac{1}{2}\lambda\boldsymbol{g}^T K_{ncs}\boldsymbol{g}. \tag{23}$$

In this study, we set the prior for $\boldsymbol{g}$ to be

$$\pi(\boldsymbol{g}) = \frac{\lambda^{\frac{K-1}{2}}}{(2\pi)^{\frac{K-1}{2}}(\mu_1\mu_2\ldots\mu_{K-1})^{\frac{1}{2}}}\exp\left(-\frac{1}{2}\lambda\boldsymbol{g}^T K_{ncs}\boldsymbol{g}\right), \tag{24}$$

where μ_1, μ_2, μ_{K-1}.are the inverses of the $N-1$ nonzero eigenvalues of K_{ncs} and λ is specified based on specific applications. During the estimation, we set $\lambda = 0.01$.

For the proposal distribution of $\boldsymbol{g}$, we set it to be.

$$\boldsymbol{g}^*|\,\boldsymbol{g}^{(i-1)} \sim MVN(g^{(i-1)}, \Sigma), \tag{25}$$

where $\Sigma = \sigma_e^2 K^-$ and K^- is the generalized inverse of K_{ncs}. The constant σ_e^2 is specified based on specific applications.

4.2. Simulation results

In this subsection, we conducted a simulation experiment to check the performance of the semiparametric SV model. The true data generating process (DGP) contains a nonlinear transition equation as follows:

Table 1
Simulation results under different values of φ.

φ	True DGP				Semiparametric SV model			
	0.7	0.8	0.9	0.95	0.7	0.8	0.9	0.95
In-sample								
Log-Likelihood	3098.8	3195.2	3041.8	3233.8	3099.1	3199.4	3043.2	3235.4
MSE: log-vol.	0.2225	0.2826	0.2421	0.3322	0.2327	0.2889	0.2383	0.3448
Out-of-sample								
Log-Likelihood	631.4	600.6	550.4	590.7	631.0	598.6	551.5	590.8
MSE: log-vol.	0.2172	0.2254	0.2361	0.2481	0.2375	0.2562	0.2497	0.2610

Table 2
Monte Carlo simulation results.

	DGP	Basic SV model	Semiparametric SV model
Parameter part			
α	0.10	0.0099686	0.0099692
[rmse]		0.0002937	0.0002884
[std]		0.0002935	0.0002882
β	0.90	0.9000392	0.9000620
[rmse]		0.0108271	0.0108492
[std]		0.0108816	0.0109037
σ	0.25	0.2339469	0.2433815
[rmse]		0.0213419	0.0177476
[std]		0.0141342	0.0165503
Nonparametric part			
IMSE: m_h		0.014620	0.009115
In-sample			
Log-Likelihood	3179.368555	3171.916656	3179.372788
MSE: log-vol	0.212773	0.249075	0.220037
Out-of-sample			
Log-Likelihood	633.135728	630.925321	632.118910
MSE: log-vol	0.210195	0.248578	0.228179

$$\begin{aligned} y_t &= \alpha + \beta x_t + \exp\left(\frac{h_t}{2}\right) v_t \quad , \\ h_{t+1} &= \left(-10 + 1.5 * cdf\left(\frac{h_t - (-10 + 0.75)}{0.5 * 0.25 \Big/ \sqrt{1-\varphi^2}}\right)\right)(1-\varphi) + \varphi h_t + 0.25 w_t \quad , \\ & 0 < \varphi < 1, \end{aligned} \tag{26}$$

where $cdf(.)$ is the cumulative distribution function of the standard normal distribution and x_t is sampled from a uniform distribution $U[-0.05, 0.05]$.

Under the specified DGP, the log-volatility has a mean value close to -9.25. Moreover, different values of φ can be selected to adjust the degree of nonlinearity in the transition equation. The slope of the transition equation is close to φ when the independent variable is far from -9.25. Under some values of φ, the slope of the transition equation is even greater than 1 when the independent variable is close to -9.25. When φ is close to 1, the transition equation is close to an $AR(1)$ process.

As for the other parameters, we set α to be 0.1, β to be 0.9, in-sample size to be 1,000, out-of-sample size to be 200, particle numbers M to be 1,000, knots number K to be 6, burn-in iterations B to be 5,000, and MCMC iterations N to be 5000.

Once we know the true DGP model, we can use the particle filter method to calculate the likelihood and estimate the log-volatility. Therefore, we can compare the results under the true DGP and the semiparametric SV model to evaluate the fitting performance of the latter.

Table 1 presents the log-likelihood and mean squared error of log-volatility under the true DGP and semiparametric SV models. For the same value of φ, the log-likelihoods and mean squared errors of log-volatility under the true DGP and semiparametric SV model are quite close for both in-sample and out-of-sample data. Therefore, the flexible semiparametric SV model can provide quite accurate estimation for the nonlinear relation in the transition equation.

Further, for $\varphi = 0.7$, we do 100 replications to evaluate the accuracy and robustness of the estimates. The in-sample size is 1,000, out-of-sample size is 200, number of particles M is 1,000, number of knots K is 6, burn-in iterations B is 3,000, and MCMC iterations N is 3000. For the parametric part, we used the mean squared error as the evaluation criterion. For the nonparametric part in the transition equation, following Ai and Chen (2003), we use the integrated mean squared error (IMSE) to evaluate the nonparametric estimates. The IMSE is approximated using the following discrete equation:

$$IMSE = 0.01 \sum_{j=0}^{\frac{h_{high}-h_{low}}{0.01}} \left[m_h(h_{low} + 0.01j) - \widehat{m}_h(h_{low} + 0.01j) \right]^2. \tag{27}$$

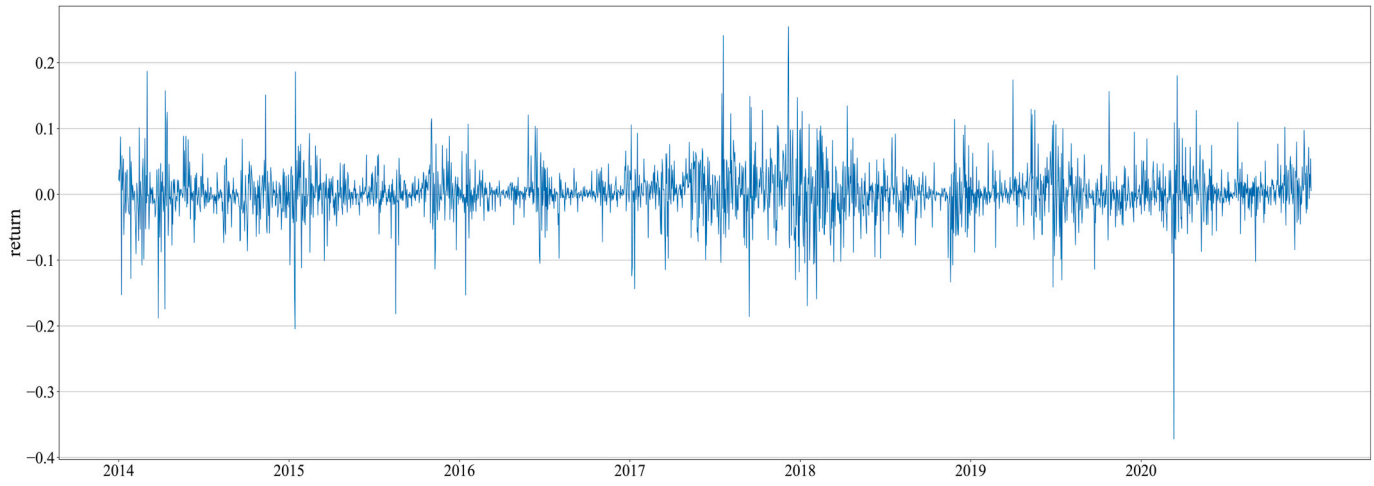


Fig. 1. Time series plot of the daily return of Bitcoin.

Table 3
Estimation results for Bitcoin return data.

	Basic SV model	Semiparametric SV model
Parameter		
α rowhead	0.00112	0.00111
σ rowhead	0.72764	0.74640
in-sample		
Log-Likelihood	5318.74332	5319.75686
Out-of-sample		
Log-Likelihood	12.50795	14.98370

parts.

Moreover, considering the log-likelihood and mean squared error of log-volatility, the semiparametric SV model performs better for both in-sample and out-of-sample data than the basic SV model. The log-likelihood and mean squared error of log-volatility in the semiparametric SV model are closer to those calculated by the true DGP.

5. Empirical applications

5.1. Empirical application to bitcoin

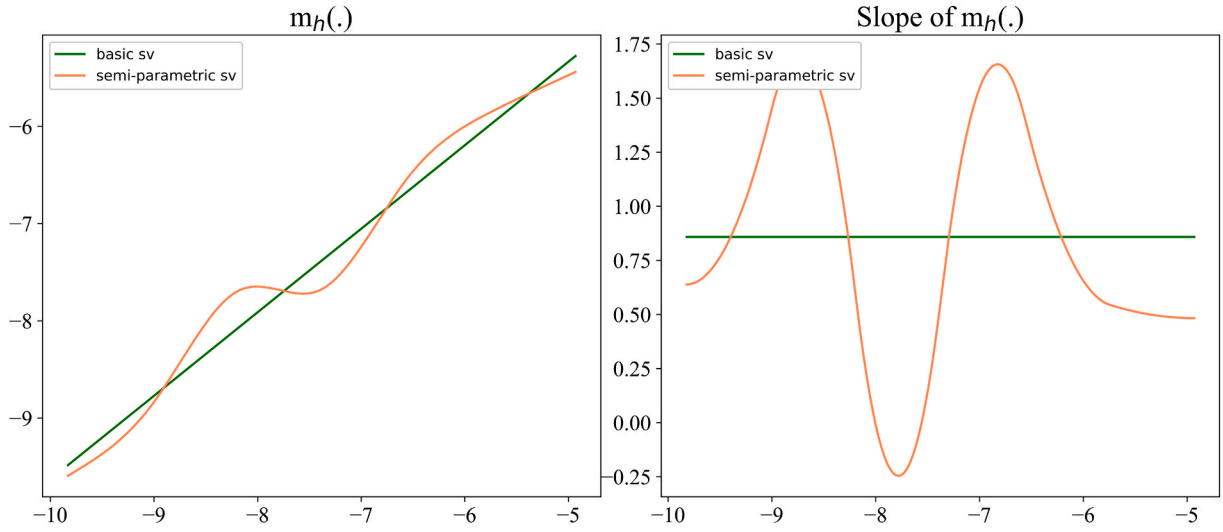


Fig. 2. Estimates of $\boldsymbol{m_h}(.)$ (left) and the slope of $\boldsymbol{m_h}(.)$ (right) for the Bitcoin daily return data.

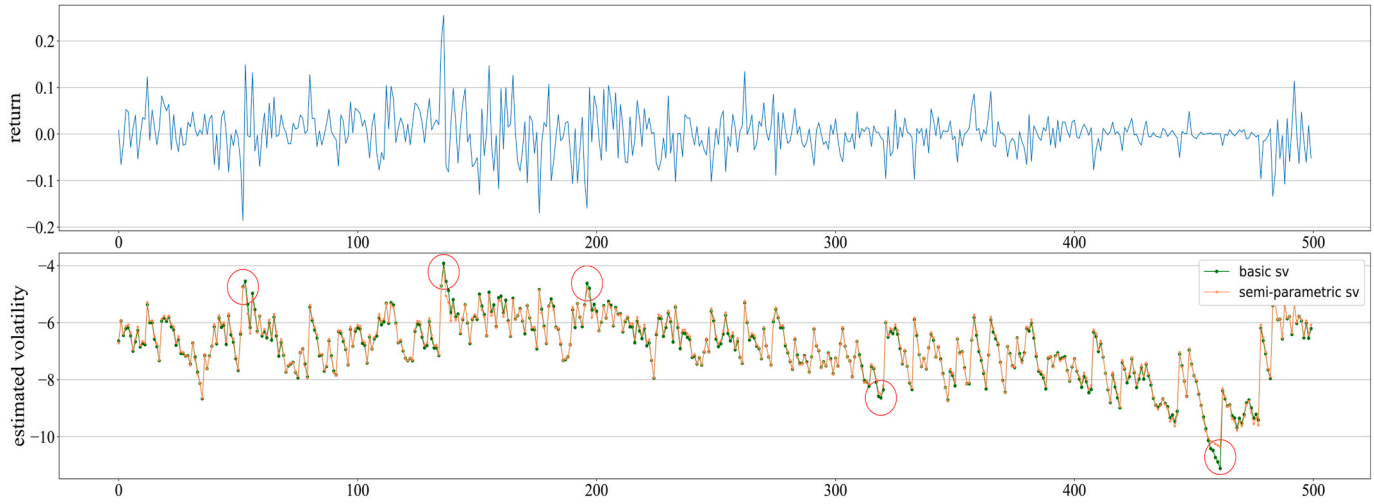


Fig. 3. Estimated log-volatility for Bitcoin return data: basic SV model and semiparametric SV model.

In each replication, we set the h_{low} and h_{high} to be 5% and 95% quantile of the distribution of h_t, respectively, and then use the posterior mean of the parameter vector $\boldsymbol{g}$ to calculate the IMSE of $\widehat{m}_h$. When 100 replications are complete, we calculate the mean IMSE across the 100 replications.

The replication results based on basic SV and semiparametric SV models are listed in Table 2. For the estimates of α and β, they are close and accurate in both the basic and semiparametric SV models. However, the semiparametric SV model can estimate σ closer to the true value and provide lower IMSE of m_h. Therefore, the semiparametric SV model can provide a better estimate for both the parametric and nonparametric

We applied our model to fit the daily return of Bitcoin from Jan 1, 2014, to Dec 31, 2020, with 2557 observations. The data resource is public and can be downloaded from https://history.btc126.com/data/btc/2012/. Bitcoin was invented in 2008 under the pseudonym Satoshi Nakamoto. The currency was first used in 2009, when its implementation was made available as open-source software. As a global cryptocurrency, the price of Bitcoin is highly volatile and can exert an impact on other financial markets through various channels such as speculation, transaction, portfolio construction, and so on. The time series plot of the Bitcoin return data we used is shown in Fig. 1. The mean and standard deviation (in percentage value) of these returns are

Table 4
Ljung–Box Q statistics for the original and filtered Bitcoin returns.

Statistics	Original data	Basic SV model	Semiparametric SV model
$Q^2(5)$	121.7306	24.3161	17.2290
$Q^2(10)$	171.6732	31.1865	27.1811

a subsample of 500 observations for plotting. It can be observed that the semiparametric SV model is smoother and adjusts more flexibly in response to volatility shocks than the basic SV model. When the volatility is extremely high (low) in specific circle areas, the basic SV model tends to overestimate (or underestimate) the volatility due to the high persistence resulting from the large constant value of the estimated

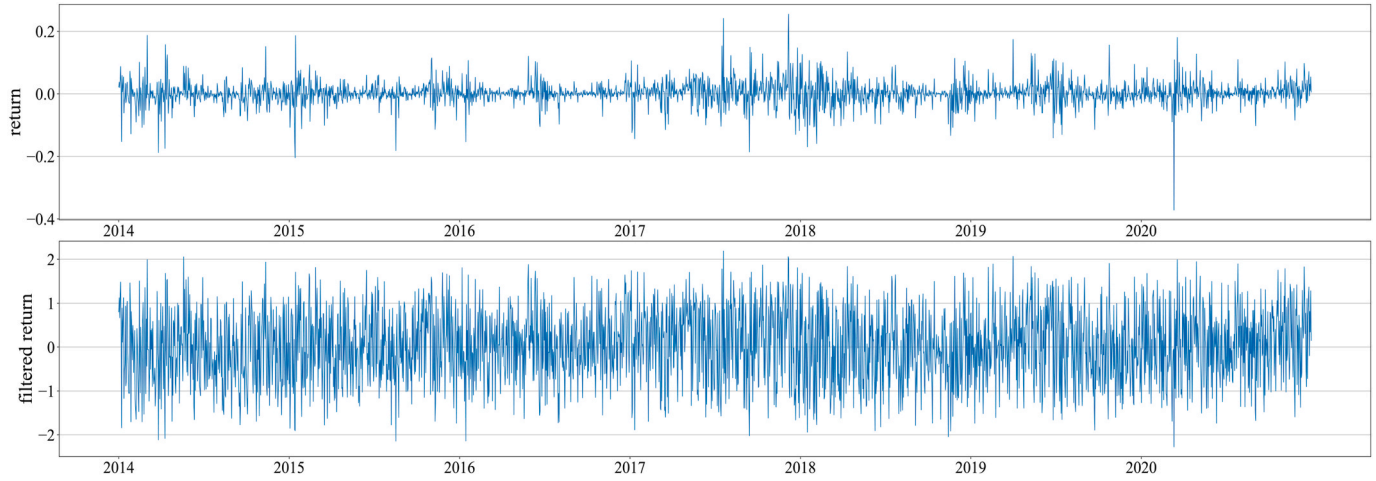


Fig. 4. Time series of the original and filtered observations for Bitcoin daily return data.

0.00215 and 0.0385, respectively. As expected, the standard deviation is much larger than the mean and the volatility is time-varying, exhibiting some degree of clustering.

To apply our model and method to Bitcoin returns, we set the number of particles M to be 1,000, number of knots K to be 6, burn-in iterations B to be 10,000, and MCMC iterations N to be 20,000. The estimation results for the parameters are listed in Table 3. The estimated values for α based on the two models are quite close. Conversely, for σ, the basic SV and semiparametric SV give estimates of 0.72764 and 0.74640, respectively. To compare the data-fitting performance of the two models, we reported the log-likelihood at the estimated value of parameters. The results show that the semiparametric SV model outperforms the basic SV model both in-sample and out-of-sample.[1]

Fig. 2 shows the shape of the cubic spline function for the nonparametric part. In the left graph, the orange line and the green line show the estimates for the $m_h(.)$ for semiparametric SV and basic SV models, respectively. The right graph exhibits the slope of $m_h(.)$. The left graph shows that the estimated function $m_h(.)$ from the semiparametric SV model is nonlinear instead of being linear as assumed in the basic SV model.

As shown in the right graph in Fig. 2, the slope of the $m_h(.)$ under semiparametric SV model varies greatly (orange line), being against the linearity assumption in the basic SV model (green line). In the middle of this interval (with a medium log-volatility), the slope of $m_h(.)$ achieves the minimum of ~ -0.25, while at the two peaks of the slope curve, the slope of $m_h(.)$ achieves the maximum of ~ 1.75. In addition, the slopes of $m_h(.)$ near the endpoints of the interval (with a low or high log-volatility) decrease to be <0.75. Therefore, the estimation result of $m_h(.)$ under semiparametric SV model confirms the necessity of using a more flexible nonparametric specification of the transition equation when fitting time-varying volatility using real data.

Fig. 3 exhibits the time series of observations and estimated log-volatility based on the basic SV (green dots) and semiparametric SV (orange dots) models, respectively. To ensure figure clarity, we selected $AR(1)$ autocorrelation coefficient. Conversely, because the semiparametric SV model has a nonlinear transition function with much lower autocorrelation coefficients at extreme volatilities than the basic SV model (right graph in Fig. 2), its estimated volatility adjusts more quickly after volatility shocks.

Table 4 lists the Ljung–Box Q test statistics (with 5 lag and 10 lag) for the original Bitcoin return data and filtered return data using the basic and semiparametric SV models. The filtered returns have much smaller Q statistics than those of the original data, indicating that the filtered returns have a much smaller variance autocorrelation than the original data. Comparing the basic and semiparametric SV models, the filtering based on the semiparametric SV model is more effective.

Finally, to visualize the filtering effect, we compared the time series of the original and filtered Bitcoin return data based on the semiparametric SV model (Fig. 4). The semiparametric SV model captures the volatility clustering feature of the original data, and the filtered returns no longer exhibit an obvious clustering phenomenon.

5.2. Empirical application to the convertible bond

We applied our model to fit the hourly return of a convertible bond for Shanghai Modern Pharmaceutical Company Limited. The sample period is from May 10, 2019 to Feb 14, 2020, with 748 observations. The data were obtained from the Wind database. A convertible bond is a kind of bond that contains an option to convert to stock at a convertible price. Therefore, with this option, its pricing principle varies according to the relative motion of the stock and convertible prices. The convertible bond behaves like an ordinary bond, option, or stock when the stock price is lower than, close to, or higher than the convertible price, respectively. Hence, we expect that the log-volatility transition equation for convertible bonds exhibits some nonlinearity.

The time series plot of the convertible bond return data we used is shown in Fig. 5. The mean and standard deviation (in percentage value) for these returns are 0.00016 and 0.0039, respectively. The standard deviation is much larger than the mean and the volatility shows evident clustering.

To apply our model and method to convertible bond returns, we set the number of particles M to be 1,000, number of knots K to be 6, burn-in

[1] The in-sample period is from Jan 1, 2014, to Dec 31, 2020. The out-of-sample period is from Jan 1, 2021, to Jan 12, 2021.

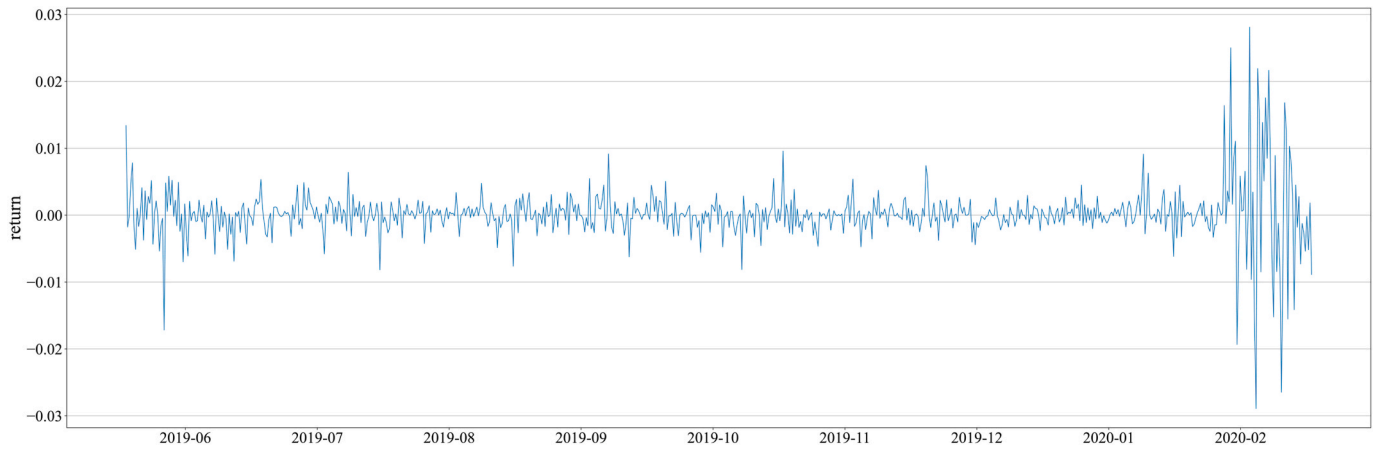


Fig. 5. Time series plot of the hourly return of convertible bond.

Table 5
Estimation results for the convertible bond return data.

	Basic SV model	Semiparametric SV model
parameter		
α rowhead	0.00009	0.00010
σ rowhead	0.83918	0.95920
in-sample		
Log-Likelihood	3461.06648	3480.84694
Out-of-sample		
Log-Likelihood	51.60532	52.67847

be a nonlinear function instead of being linear as in the basic SV model. In the right graph in Fig. 6, the slope of $m_h(.)$ estimated by the semiparametric SV model varies across different values of log-volatility. For instance, in the middle of this interval (with a medium log-volatility), the slope of $m_h(.)$ achieves the minimum value of ~ -0.75, while at the peak of the slope curve, the slope of $m_h(.)$ achieves the maximum value of ~ 2.75. Moreover, the slopes of $m_h(.)$ near endpoints of the interval (with a low or high log-volatility) decrease to be ~ 0.5.

Fig. 7 illustrates the time series of observations of convertible bond returns and the estimated log-volatility based on the basic SV (green

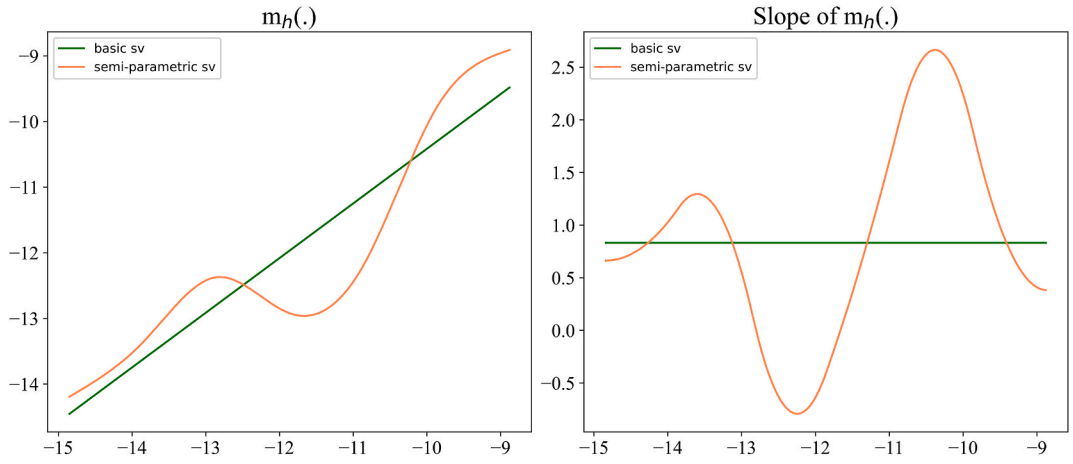


Fig. 6. Estimates of $m_h(.)$ (left) and slope of $m_h(.)$ (right) for hourly return of the convertible bond.

iterations B to be 10,000, and MCMC iterations N to be 20,000. The estimation results for the parameters are listed in Table 5. The estimated α values based on the two models are quite close, while for σ, the basic SV and semiparametric SV models give estimates at 0.83918 and 0.95920, respectively. Comparing the log-likelihood evaluated at the estimated values of parameters, we can see that the semiparametric SV model is better than the basic SV model both in-sample and out-of-sample.[2]

For the nonparametric estimation results, we plot the estimated cubic spline function in Fig. 6.

Similar to the Bitcoin return data shown in Fig. 2, the results shown in Fig. 6 confirm that for the convertible bond return data, $m_h(.)$ should

dots) and semiparametric SV (orange dots) models, respectively. To ensure figure clarity, we selected a subsample of 500 observations for plotting. Similar to the Bitcoin return data shown in Figs. 3 and 7 provides more distinct evidence that the semiparametric SV model adjusts the estimated volatility more flexibly than the basic SV model following volatility shocks. Some areas are circled as examples in Fig. 7. The reason for this difference between basic SV and semiparametric SV models has been explained in Section 5.1.

Table 6 lists the Ljung–Box Q test statistics (with 5 lag and 10 lag) for the original convertible bond return data and filtered data using the basic SV and semiparametric SV models. We can see that the filtered volatility has a much lower variance autocorrelation than the original data, and the semiparametric SV model has a better filtering effect than the basic SV model.

Fig. 8 shows the time series of the original convertible bond return

[2] The in-sample period is hourly data from May 10, 2019 to Feb 14, 2020. The out-of-sample period is hourly data from Feb 15, 2020 to Feb 19, 2020.

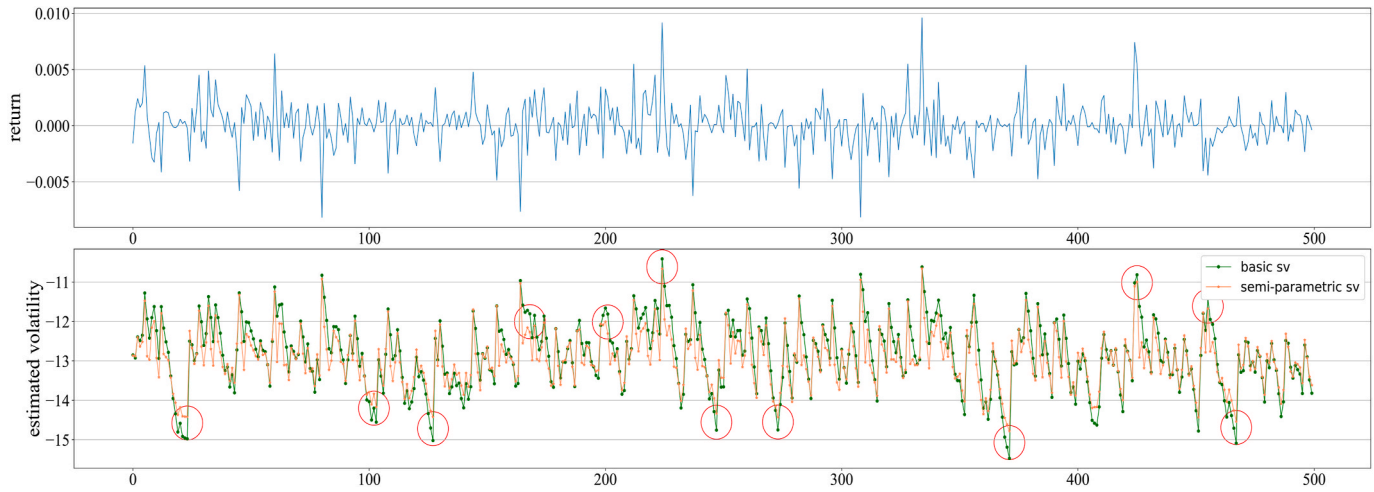


Fig. 7. Estimated log-volatility for the convertible bond return data: basic SV and semiparametric SV models.

Table 6
Ljung–Box Q statistics for the original and filtered convertible bond returns.

Statistics	Original data	Basic SV model	Semiparametric SV model
$Q^2(5)$	439.6238	28.3689	18.0646
$Q^2(10)$	724.2714	47.0721	31.3222

data and filtered ones based on the semiparametric SV model. After filtering, the returns no longer exhibit an obvious clustering phenomenon.

6. Conclusion

In this study, we proposed a flexible semiparametric SV model using a natural cubic spline function in the transition equation. Compared to the basic SV model, the nonparametric specification of the transition equation provides more generality and flexibility, thus exhibiting better empirical performance in terms of data fitting. To estimate the proposed semiparametric SV model, we applied the SMC algorithm and used the PMCMC method.

The simulation results indicate that the PMCMC method can provide an effective estimate of the semiparametric SV model, and this semiparametric SV model can fit the nonlinearity of the transition equation well. Compared with the basic SV model, the semiparametric SV model exhibits smaller parameter estimation bias, smaller root mean squared error, and smaller IMSE. Empirical applications to Bitcoin and convertible bond return data indicate that there is actually high nonlinearity in the transition equation of their log-volatility. Taking nonlinearity into account can improve the data-fitting performance of SV models. This highlights the necessity of using our semiparametric SV model to fit the real data.

Finally, because we are focusing on the nonparametric extension of the transition equation in the SV model, we assume the Gaussian distribution of innovation for simplicity in this study. However, our model can be conveniently modified to accommodate other variants of SV models assuming more general innovation distributions, such as the SV model with non-Gaussian innovations (Jensen and Maheu, 2010; Langrock et al., 2015) and the SV model with the leverage effect (Yu, 2005). Moreover, the estimation method used herein is also applicable for these extensions as long as the density functions are available. It would be worthwhile to test whether these combinations could further improve the performance of SV models in a future study.

Funding details

Wang gratefully acknowledges the financial support of the Chinese Natural Science Fund (No.72203151) and the support by Capital University of Economics and Business (No. XRZ2022037).

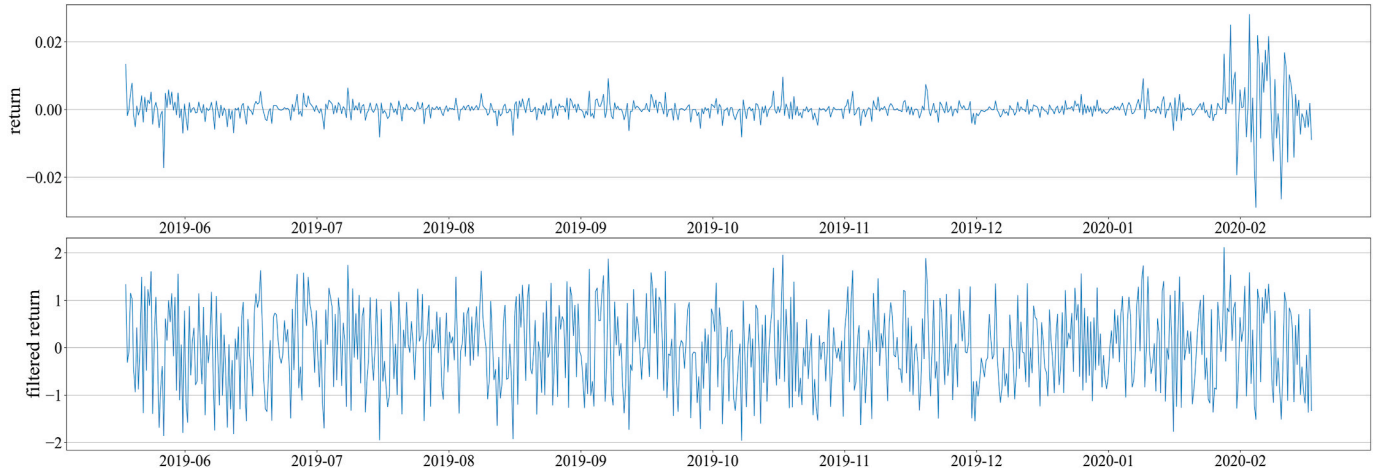


Fig. 8. Time series of the original and filtered observations for the hourly return of the convertible bond.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

We would like to thank Sushanta Mallick (the editor), Yong Li (the guest editor) and two referees for their valuable comments, which helped a lot to improve the quality of our paper.

References

Ai, C., Chen, X., 2003. Efficient estimation of models with conditional moment restrictions containing unknown functions. Econometrica 71 (6), 1795–1843.
Andrieu, C., Doucet, A., Holenstein, R., 2010. Particle Markov chain Monte Carlo methods. J. Roy. Stat. Soc. B Stat. Methodol. 72 (3), 269–342.
Baek, C., Elbeck, M., 2015. Bitcoins as an investment or speculative vehicle? A first look. Appl. Econ. Lett. 22 (1), 30–34.
Chen, J.P., Xiong, X., Zhu, J., Zhu, X.N., 2017. Asset prices and economic fluctuations: the implications of stochastic volatility. Econ. Modell. 64, 128–140.
Cross, J.L., Hou, C.H., Trinh, K., 2021. Returns, volatility and the cryptocurrency bubble of 2017–18. Econ. Modell. 104, 105643.
Delatola, E.I., Griffin, J.E., 2011. Bayesian nonparametric modelling of the return distribution with stochastic volatility. Bayesian Analysis 6 (4), 901–926.
Dyhrberg, A.H., 2016. Bitcoin, gold and the dollar – a GARCH volatility analysis. Finance Res. Lett. 16, 85–92.
Green, P.J., Silverman, B.W., 1993. Nonparametric Regression and Generalized Linear Models: a Roughness Penalty Approach. Chapman and Hall/CRC.
Hafner, C.M., 2020. Testing for bubbles in cryptocurrencies with time-varying volatility. J. Financ. Econom. 18 (2), 233–249.
Harvey, A.C., Ruiz, E., Shephard, N., 1994. Multivariate stochastic variance models. Rev. Econ. Stud. 61, 247–264.
He, X.J., Lin, S., 2023. A new nonlinear stochastic volatility model with regime switching stochastic mean reversion and its applications to option pricing. Expert Syst. Appl. 212, 118742.
Hu, S.W., Poskitt, D.S., Zhang, X.B., 2021. Bayesian estimation for a semiparametric nonlinear volatility model. Econ. Modell. 98, 361–370.
Huang, A.Y., 2015. Value at risk estimation by threshold stochastic volatility model. Appl. Econ. 47 (45), 4884–4900.
Hull, J., White, A., 1987. The pricing of options on assets with stochastic volatilities. J. Finance 42 (2), 281–300.
Jacquier, E., Polson, N., Rossi, P.E., 1994. Bayesian analysis of stochastic volatility models. J. Bus. Econ. Stat. 12, 69–87.
Jensen, M.J., Maheu, J.M., 2010. Bayesian semiparametric stochastic volatility modeling. J. Econom. 157 (2), 306–316.
Kim, S., Shephard, N., Chib, S., 1998. Stochastic volatility: likelihood inference and comparison with ARCH models. Rev. Econ. Stud. 63, 361–393.
Langrock, R., Michelot, T., Sohn, A., et al., 2015. Semiparametric stochastic volatility modelling using penalized splines. Comput. Stat. 30 (2), 517–537.
Li, Y., Chong, T.T.L., Zhang, J., 2012. Testing for a unit root in the presence of stochastic volatility and leverage effect. Econ. Modell. 29 (5), 2035–2038.
Li, Y., Zhang, M.Z., Zhang, Y.H., 2022. Sequential Bayesian bandwidth selection for multivariate kernel regression with applications. Econ. Modell. 112, 105859.
Pan, Q., Li, Y., 2013. Testing volatility persistence on Markov switching stochastic volatility models. Econ. Modell. 35, 45–50.
Sandmann, G., Koopman, S.J., 1998. Estimation of stochastic volatility models via Monte Carlo maximum likelihood. J. Econom. 87, 271–301.
Tiwari, A.K., Kumar, S., Pathak, R., 2019. Modelling the dynamics of Bitcoin and Litecoin: GARCH versus stochastic volatility models. Appl. Econ. 51 (37), 4073–4082.
Ulm, M., Hambuckers, J., 2022. Do interest rate differentials drive the volatility of exchange rates? Evidence from an extended stochastic volatility model. J. Empir. Finance 65, 125–148.
Xu, W.D., Wu, C.F., Li, H.Y., 2011. Foreign equity option pricing under stochastic volatility model with double jumps. Econ. Modell. 28 (4), 1857–1863.
Yu, J., 2002. Forecasting volatility in the New Zealand stock market. Appl. Financ. Econ. 12, 193–202.
Yu, J., 2005. On leverage in a stochastic volatility model. J. Econom. 127 (2), 165–178.
Yu, J., Yang, Z.L., Zhang, X.B., 2006. A class of nonlinear stochastic volatility models and its implications for pricing currency options. Comput. Stat. Data Anal. 51 (4), 2218–2231.
Yu, J., 2012. A semiparametric stochastic volatility model. J. Econom. 167 (2), 473–482.
Zeng, X.D., Taksar, M., 2013. A stochastic volatility model and optimal portfolio selection. Quant. Finance 13 (10), 1547–1558.